

Predicting the Near-Weekend Ticket Sales Using Web-based External Factors and Box-office Data

Seonghoon Moon, Suman Bae, Songkuk Kim

Yonsei Institute of Convergence Technology

School of Integrated Technology

Yonsei University

Incheon, Korea

Email: {seonghoon.m, befree13, songkuk}@yonsei.ac.kr

Abstract—Posting online reviews and rating their satisfaction on purchased products has become an increasingly popular way to share the information for anonymous candidates who has interest in purchasing the product. In addition, people leave their interests and near-future purchasing plan on the web such as search history and search query volume. From this phenomenon, the prediction of sales performance is possible in many products by mining the data sets which are left on the web by consumers' online activities. In this paper, we focused on the movie ticket sales which word-of-mouth effect is prominent, and our goal is to forecast the sales performance of the near-weekend using box-office data and external factors such as online reviews, star ratings and search volume. For this work, we gather 1.7 million online reviews and movie ratings, and we also gather the daily search volume of movies' title for past three years. Using machine learning techniques and linear modeling, we develop a model for high-accuracy predicting of ticket sales on near-future. We also analyze a relationship between ticket sales performance on weekends and box-office data, online reviews, star ratings, and search volume. Through this work, we support to decide the ideal number of screens for a given weekend, thus it contributes to a substantial increase in the rate of profit on movie markets.

I. INTRODUCTION

With the development of Web 2.0 technologies and the diffusion of mobile Internet access, people can easily obtain and share the information through the Internet. The rise of social networking services (SNS) also accelerates the sharing of information and their opinion via the Internet. This phenomenon leads to significant changes in the pattern that people make a decision to purchase products. Internet users get information of products through various sources such as online communities, newsgroups and SNS before the new products is released. Then they use search engines in order to collect details in products. In this process, the customer makes decisions on products and purchases them through e-commerce services. After the purchase, the consumer spends time to write online reviews on multiple platforms for anonymous candidates who has interest in purchasing the product. This series of patterns mentioned above, would not be a new phenomenon anymore.

From this pattern, the prediction of sales performance is possible in many products by mining the data sets which is left on the web by consumers' online activities. Based on these ideas, numerous studies have shown that search patterns and online reviews have a close relationship with the product's sales performance [1], [2]. Moreover, there have

been several works to forecast new product sales by web data analysis in various categories [3], [4]. For the improvement of prediction accuracy, several studies introduced prediction models combined with machine learning techniques [5], [6].

Prior studies on the predicting sales performance with web data analysis have been treated in various categories (e.g. book, appliances, video games, etc.). In this paper, we present the prediction model for ticket sales on near weekend by analyzing box-office data, online reviews, star ratings and search term volume data of the past three years. The reasons that we choose the movie domain, first of all, is that it is easy to collect the precise measured data of the past sales performance. Second, the movie domain is familiar to purchasing patterns through word-of-mouth (WOM) as people recommend movies through online reviews and star ratings [1]. Third, it is relatively simple to crawl the web data, because reviews and star ratings are gathered in a few critic reviews websites and portal services. At last, in 2012, global theatrical market is a huge size market of \$34.7 billion, and specifically the Korean film market size is \$1.3 billion, which is about 3.8 percent of the global market [7]. Thus, a high revenue growth can be expected when the cinema decides the ideal number of screens through an accurate prediction of near-future ticket sales.

However, the prediction of ticket sales is actually a much harder problem. Fig. 1 is showing the amount of normalized weekend ticket sales for three years from 2010. Generally, it can be shown that the sales performance decreased rapidly as time passed after the movie has been released. On the other hand, Fig. 2 is presenting the weekends ticket sales of two movies namely 'Sunny' and 'Last godfather' which are released in 2011. It is interesting to observe from Fig. 2 that two movies are showing different sales patterns, compared with Fig. 1 in above. The 'Sunny' has increasing sales performance caused by good star rating and viral effect. On the contrary, 'Last godfather' has abnormal decreasing sales performance. TABLE I shows the values of external factors of 'Sunny' and 'Last godfather' for a week before release and after release, that are online reviews volume, movie ratings from Naver movie service and calculated search query volume from Google Trends and Naver Trends. Comparing with the average values given below the table, the 'Sunny', unlike the average values, showed the increased movie rating as 9.57 for a week after release, and also showed significant increase of online comments volume more than six times. This can be determined that the audiences reaction to the 'Sunny' had more

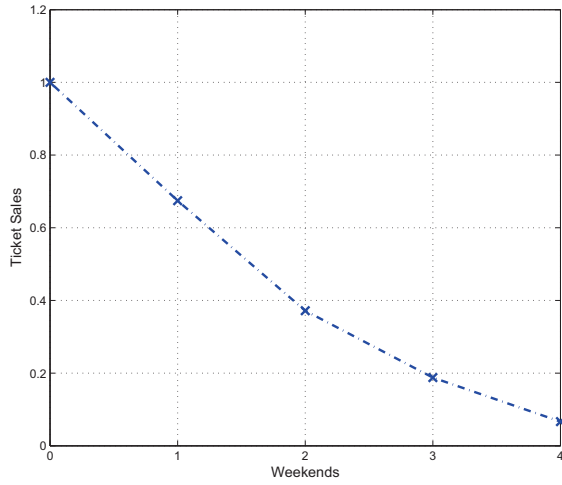


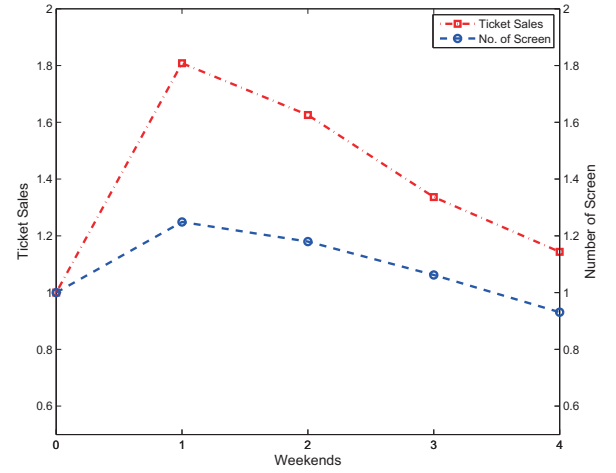
Fig. 1: An average weekly ticket sales pattern.

TABLE I: The weekly values of movie rating, online comment volume and search volume for before-release and after-release of ‘Sunny’, ‘Last godfather’ and average of movies

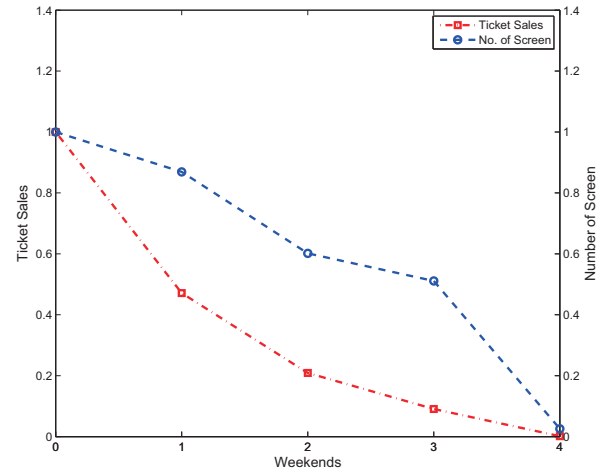
Movie	Movie ratings		Comment volume		Search volume	
	before-	after-	before-	after-	before-	after-
‘Sunny’	9.39	9.57	128	761	23.25	61.29
‘Last godfather’	9.14	6.98	1070	2778	24.90	67.50
Average	8.71	8.10	355	1347	25.78	59.50

positive than the expectation at before release. However, the ‘Last godfather’ had the significant decreased movie rating as 6.98 after release, and it is possible to assume that the negative viral was proceeded actively with the large amount of online comments volume. Moreover, we can recognize from Fig. 2 that a difference exists between ticket sales pattern and the number of screens, which could be attributed to the wrong prediction of ticket sales by a human expert. Through this study, we propose a novel system to predict ticket sales performance on the nearest weekend with high accuracy using a time-series data analysis of box-office data and external factors that reflect sales performance such as online reviews, star ratings and search query volume. Additionally, as seen in Fig. 3, predicting the ideal distribution of screen for weekends contributes to a substantial increase in the rate of profit on movie markets, due to the fact that weekend ticket sales contribute to more than half of the total revenue.

For this study, we aggregated box-office data of 110 movies released from January 2010 to October 2013, offered by the Korean Film Council. We also crawled 1,683,782 of movies’ review comments and star ratings from the movie service of “Naver”, one of the largest portal sites in Korea. Additionally, for the daily search term volume of movies’ title, we crawled Google Trends and Naver Trends data from January 2010 to October 2013. At first, we investigate empirically how the nearest weekend ticket sales are correlated with the data



(a)



(b)

Fig. 2: An example of two different ticket sales patterns. The weekends ticket sales of (a) ‘Sunny’ (b) ‘Last godfather’

sets we aggregated. Moreover, we compare the performance of the prediction model using well-known machine learning techniques such as ANN (Artificial neural network), SVM (Support vector machine) and LM (linear modeling).

In previous works, they provided experimental results of the relationship between online reviews, search term volume data and box-office performance [1], [2]. Or they proposed prediction models for total revenue of newly released movies [3], [5], [8]. Compared with the previous work above, we propose a predicting model of ticket sales for every weekend by analyzing the time-series data of movies’ internal factors and external factors. In summary, we make the following contributions:

- We present an empirical result, which maps a relation-

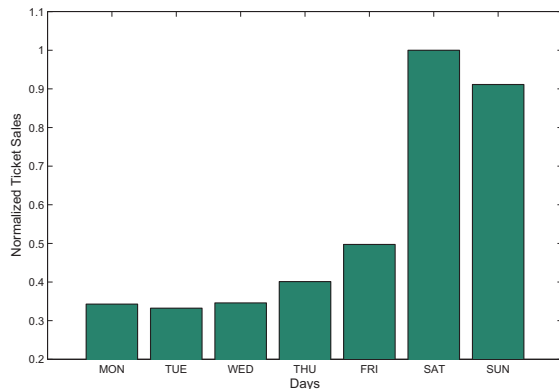


Fig. 3: The normalized value of daily ticket sales performance.

ship between ticket sales performance on weekends and box-office data, online reviews, star ratings, and search query volume.

- We develop a model for high-accuracy predicting of ticket sales on the nearest weekend using several machine learning techniques and linear modeling.
- Using proposed prediction models we analyze how much the accuracy is improved by using external factors (e.g. volume of online reviews, star ratings and search volume), which reflect ticket sales performance.
- We help to decide the ideal number of screens on weekends, and then bring a direct effect with improving profit in movie markets through this advantage.

The rest of this paper is organized as follows. Section II introduces studies related to our works. Section III explains the method of data acquisition, preprocessing and describes characteristics of data sets. Section IV gives the overall of proposed prediction system and an experiment result. The final section provides the conclusion and direction for future development.

II. RELATED WORK

In the past few years, the prediction of various product sales with data mining has emerged as an attractive theme for knowledge discovery. The movie domain is also treated in several studies with various perspective and data sets. The works are generally investigated via the revenue of newly released movies using the analysis of past box-office performance data or characteristics of movies (e.g. genre, MPAA rate, budget, director, etc.) [5], [6], [8]. In [5], [6], [8], at first, they proved the relationship of ticket sales performance and their preprocessed data sets with a correlation coefficient, and then based on data descriptions, appropriate data features are selected for prediction modeling. The works adopted various methods for modeling such as a Bayesian belief network, a decision tree and an ANN. However, previous studies, which treated only the internal data of movies, were not focused on external factors which could affect the potential customers. In addition, for performance evaluation, they set pre-defined

classes of revenue size and measured hit rate of the test movies' class using their prediction model, instead of predicting exact figures of the revenue or ticket sales. We focus to present the prediction model to forecast the nearest weekend ticket sales, not the pre-defined classes.

In recent years, for more accurate forecasting of products sales, many studies has been utilizing not only the internal factors, but also the external factors of products such as online customer reviews, news, and social networking data. These product's external factors were proved by many studies that they are highly influential to future behaviors of potential consumers. For example, in [1], [9], present experimental results that show correlated sales performance with online customer reviews and customer satisfaction data. These previous studies prove the effect of online word-of-mouth (WOM) on the movie ticket sales performance. The other method to analyze the power of online WOM, it is possible to experiment that how the product sales are correlated with amount of contents on the social networking service (SNS) such as tweets in Twitter [10], [11]. Furthermore, in [3], they investigated the problem of predicting ticket sales performance using past box-office data and sentiment information mined from online reviews.

On the other hand, a number of empirical studies have been conducted to analyze potential customer behavior by using search volume data. People get the information from multiple routes such as news, social media and commercial advertising before they decide to buy products. And then, they would like to collect more detailed information of the products or online reviews using search engines. This process is quite familiar to the consumer nowadays. Nevertheless, it is difficult to make a judgment that every people's act of the search would be connected to the act of buying, but we can understand the product's trend and consumers' interest through the search pattern. There are also studies that predict sales performance of the movie domain using the search query volume. In [2], a study predicts opening weekend box-office revenue, sales of video games, and rank of songs using search query volumes and a few features of products' characteristic. In [4], they present a modeling framework that links search volume and box-office revenue.

Our work is similar to above studies, that we also exploit the past box-office data, characteristics of movies, online reviews, star ratings and search query volume to predict the future sales performance. However, their approach mainly focuses on opening weekend sales performance or the total amount of revenue, while our method aims to predict the nearest weekend ticket sales using the time-series analysis, and then we support to making decision of ideal number of screens. Thus, compared with previous works above, our goal solves the real issues that improve the profit on the theatrical market through more accurate model using machine learning techniques.

III. DATA SET

A. Data Acquisition and Preprocessing

We collected box-office data from 2010 to October 2013 for this study. Already mentioned above, compared with other domains, the movie domain has well-organized various sales performance data on the web that are easy to

access. This advantage also applies to the Korean film market, thus we can crawl the ticket sales data from the website (<http://www.kobis.or.kr>) operated by the Korean Film Council. We configure raw data set as 110 movies that have more than 400 screens at its first release weekend. Then, we preprocess various sales performance data sets to make features for the prediction modeling.

In order to analyze the external factors other than sales performance, we collect online reviews, star ratings and search volume data of movies' title. At first, we crawl the online reviews and star ratings from the movie service operated by "Naver" the most popular portal site in Korea. The Naver movie platform is set up to add a star rating mandatory when the user writes a review comment, so even if there is no sentiment analysis of any review comment, we can quantify roughly the emotions that have been put in each review comment. We measure the reviews volume and star rating after aligning as a time series, and then process to make features about online reviews and star ratings. In order to reduce the time to process the large amount of review data, we conduct a preprocessing using the HBase that is based on the apache Hadoop project. Second, we aggregate search volume data of movies title from Google Trends and Naver Trends services. Although obtaining box-office data, online reviews and star rating are straight-forward, collecting search volume data is more challenging because of the format of data from sources. For example, the search volume that was collected from Naver Trends provide a weekly aggregate value, thus we need to use Google Trends values in addition to get daily search volume patterns. For this reason, we estimate a feature of search volume using the equation (1) as follow.

$$SV_{tot} = 0.15 * SV_{nav} + 0.15 * SV_s + 0.3 * SV_m + 0.4 * SV_t \quad (1)$$

Where SV_{nav} is the previous week search volume of Naver Trends, SV_s is the Sunday search volume of Google Trends, SV_m is the Monday search volume of the Google Trends and SV_t is the Tuesday search volume of Google Trends. And their weight values are determined by experimental method that has the most accurate prediction performance.

At last, in order to consider the side effect of new release competition movies on prediction target weekend, we define a competition feature [12]. For calculation of the competition feature, we make a formula using expectation values of movies' characteristic such as directors, starring actors from their past movies and number of screen at launch day which reflect the expectations of the potential audience. So our formula of competition feature is like this.

$$CS = 0.5 * Screen_{cp} + 0.25 * DirectorValue + 0.25 * PlayerValue \quad (2)$$

Where CS is the competition score, $Screen_{cp}$ is the competition movies' normalized number of screens which releases following week, $DirectorValue$ and $PlayerValue$ are normalized quantitative value based on previously appeared movie. Then, we also decide their weight values experimentally which have the best prediction performance.

B. Description of Data

1) *Box-office Data*: We gathered the following features from the box-office data and aligned the data on a weekly basis: Week number after the opening week, ticket sales of the

TABLE II: Correlation coefficient between the box-office data and ticket sales of each weekend

Factor	Correlation coefficient
Week number	-0.4027
Ticket sales of the previous weekend	0.7926
Number of screens for weekdays	0.8206
Seat share of the previous weekend	0.6441
Seats per screen for weekdays	0.7101

TABLE III: Correlation coefficient between the external factors and ticket sales of each weekend

Factor	Correlation coefficient
Star rating	0.2156
Comment volume of previous weekend	0.6609
Search volume	0.4776
Competition	-0.5994

previous weekend, seat share of the previous weekend, number of screens for weekdays, seats per screen for weekdays. Our goal is to determine the number of screens throughout the predicted ticket sales using our proposed prediction model. The South Korean film market determines the number of weekend screens on Wednesday to sell tickets in advance. With reference to this fact, we preprocess the features of the box-office data using the data available before Wednesday. TABLE II shows the correlation between our features and ticket sales of each weekend. Week number after the opening week shows negative correlation with ticket sales of the weekend. The correlation between ticket sales of the previous weekend and the number of screens weekdays are 0.7926, 0.8206. So these features have a strong correlation with ticket sales of weekend.

2) *Online reviews, Star rates and Search Volume*: In contrast to the box-office data of movies, we acquired the following external factors using online reviews, star ratings and search volume data: movie rating, comment volume of previous weekend, search volume and competition. We acquired search volume and competition using the formula explained previously. TABLE III shows the correlation coefficient between the external factors and ticket sales of each weekend. As shown in the TABLE III, the volume of comments is more correlated with the ticket sales of the following box-office than the movie rating, and the search volume is not directly correlated with the weekend's ticket sales. High value of competition can make the ticket sales of the following weekend go down. The negative correlation value between competition and ticket sales supports this fact.

IV. EMPIRICAL STUDY

A. Prediction Model

The entire system process is depicted as Fig. 4. As shown in Fig. 4, the prediction model processed with selected features that we defined previously. In this paper, three different kinds of prediction models such as LM, SVM and ANN are used to fit the relationship between input features and the output. Firstly, we used a linear model. Linear models are assumed to be a good prediction model for forecasting the box office

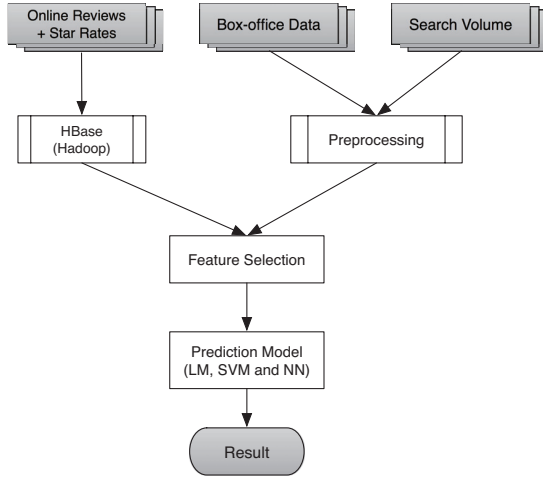


Fig. 4: An overall of the proposed prediction system.

and stock market since people assume the linear relationship between features and the output [11]. We construct the linear model as follows.

$$Tics_{wk} = w_0 + \sum_{i=1}^n w_i x_i \quad (3)$$

where $Tics_{wk}$ is the audience of next weekend and x_i is the i -th input feature. Then, we used a support vector machine's (SVM) regression model. Compared to the linear model, SVM is a non-linear model and is able to fit most non-linear models [13]. We implemented SVM in R using e1071 package and radial-basis kernel. Lastly, an artificial neural network (ANN) with a back-propagation method are used [14]. An ANN depicts action of neurons in our brain and is a powerful nonlinear regression model. We built ANN using the RSNNS package in R. The number of hidden layers and the hidden nodes in neural networks are important in determining performance. We used a multilayer perceptron (MLP) with 2 hidden layers since it gives a better result than using 1 hidden layer, and assigned five hidden nodes to each layer. It is important to set a reliable experimental design not only for getting the true accuracy of a classifier, but also for comparing among different types of prediction models. In this paper, we used k -fold cross validation. k -fold cross validation is a popular technique to get the true accuracy of a prediction model. In k -fold cross validation, the original data set (D) is randomly split into k mutually exclusive subsets of approximately equal size. Single subsamples of the data set are used for test data while $k-1$ subsamples are used for training data. This classification model is trained and tested k times while changing the test data each time ($t \in \{1, 2, \dots, k\}$). We performed 10-fold cross validation here.

B. Experiment Result

We compared the experimental result using only box-office features versus box-office features with external features, and LM, SVM and NN prediction model are applied. We conduct the experiment using 358 datasets from all the 110 movies

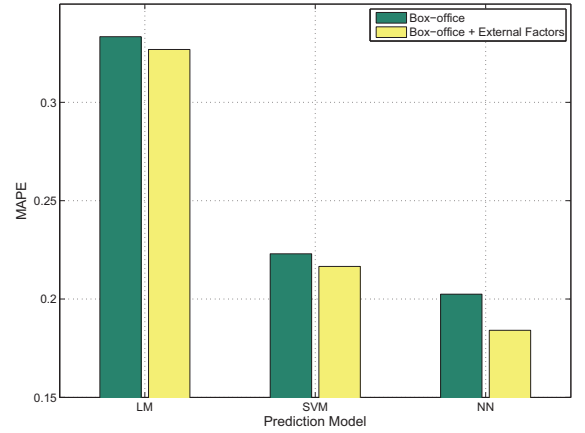


Fig. 5: Comparing the prediction performance between box-office features only versus box-office features with external factors.

and our prediction results are shown as Fig. 5. We used mean absolute percentage error (MAPE) as the evaluation criteria.

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|Pred_i - Actu_i|}{Actu_i} \quad (4)$$

Where n is the total number of the dataset, 358, and $Pred_i$ is the predicted value and $Actu_i$ is the actual ticket sales of that weekend. As shown in Fig. 5, SVM and NN prediction model shows better results than LM regardless of the features. When only the box-office data is used as a feature, MAPE of NN is 0.2025 and SVM is 0.2230 and LM is 0.3334. In the NN model showing best prediction result, the MAPE using box-office data and external features is 0.1841. Therefore about 9% of performance improvement is achieved using box-office data and external factors compared to only using box-office data. Prediction performance per week according to various prediction models is shown in Fig. 6. NN prediction shows steady performance as each week goes by, while SVM and NN prediction models show high MAPE as each week goes by. Finally, prediction results of 'Sunny' and 'Last Godfather', in which the number of screens is not determined well, are shown in Fig. 7. MAPE of 'Sunny' is 0.0961 and 'Last Godfather' is 0.1176.

V. CONCLUSION

In this paper, our goal is to forecast the sales performance of the near future using box-office data and external factor data such as online reviews, movie ratings and search volume. Especially, we focused on the movie ticket sales which WOM effect is prominent. Our prediction model forecasts the ticket sales of each weekend and can determine the ideal screen number based on our predicted ticket sales. We gather 1.7 million online reviews and movie ratings from South Korea's biggest movie site, Naver movie service. We also gather search volume from Google Trends and Naver Trends. We use these data as external factors, which affects the ticket

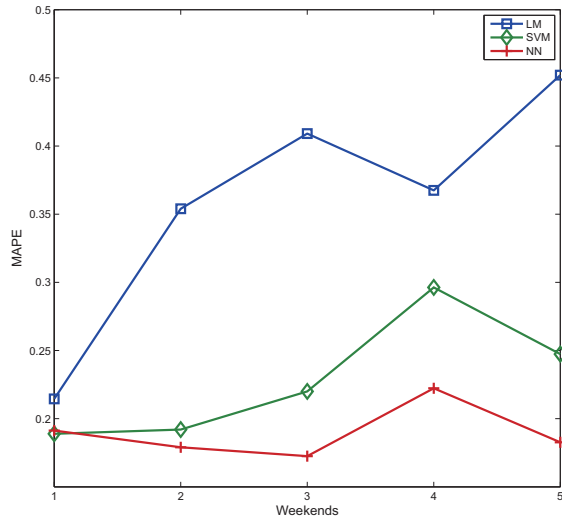
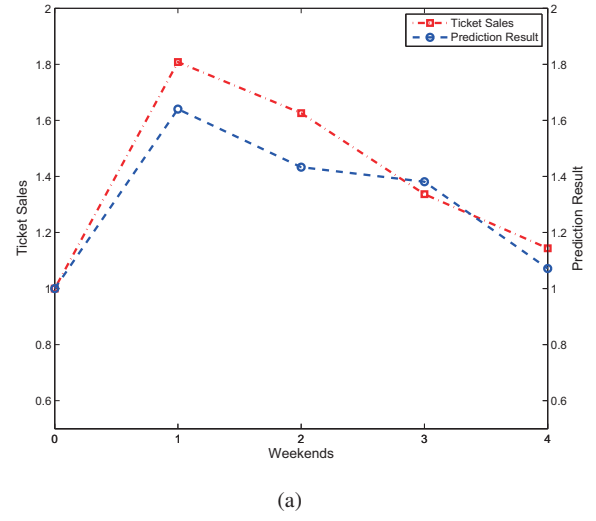


Fig. 6: The prediction performance per week according to various prediction models.

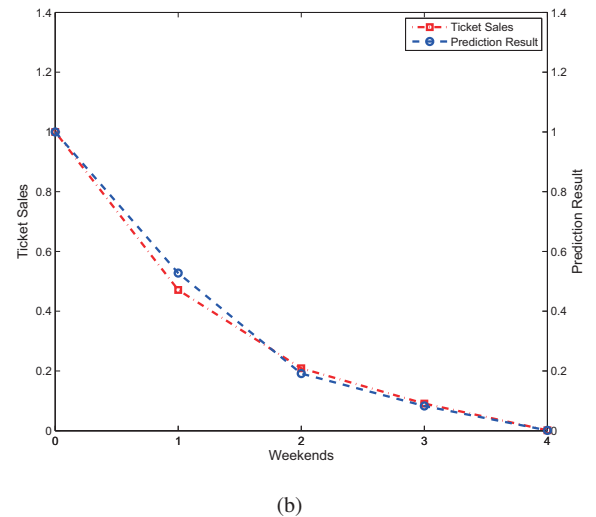
sales of movies. Together with box-office data, external factor data is used as a feature for our prediction model. We made a prediction model using LM, SVM and NN and compare the result when only box-office data is used and when box-office data with external factor data are used. In NN, which shows best performance among them, we can get about 9% improvements when external factor data such as volume of comments, movie ratings, search volume and competition are used.

The previous studies about predicting the success of box office is mainly focused on forecasting the total revenue that a movie gained using features such as the budget of movie, number of screens in the first week of release, microblog information before its release date and the information about movie. From an application perspective, however, forecasting following a weekend's ticket sales is more practical since we can flexibly determine the weekend's number of screen based on the predicted ticket sales. The ticket sales of weekends occupy about half of the total ticket sales, so determining the proper number of screens for a given weekend is the most important to maximize the sales of a theater. We demonstrate the prediction model of weekend ticket sales in this paper and show that we can get more accurate predictions using the information gained from the web, such as comment data and search volume.

According to our work, there are several things to cover in our future work. Firstly, we only gathered comment data from South Korea's biggest movie site, but if we include microblog data such as tweets in Twitter, our prediction methods may give a more accurate result. And the comment data from Naver's movie site may include advertising comments. Removing these advertising comments or separating these advertising comments may show interesting results.



(a)



(b)

Fig. 7: An example of the prediction results of (a) 'Sunny' and (b) 'Last Godfather'

ACKNOWLEDGMENT

This research was supported by the MSIP(Ministry of Science, ICT and Future Planning), Korea, under the IT Con-silience Creative Program (NIPA-2014-H0201-14-1001) super-vised by the NIPA(National IT Industry Promotion Agency), in part by Yonsei University Research fund.

REFERENCES

- [1] W. Duan, B. Gu, and A. B. Whinston, "Do online reviews matter ? An empirical investigation of panel data," vol. 45, pp. 1007–1016, 2008.
- [2] S. Goel, J. M. Hofman, S. Lahaie, D. M. Pennock, and D. J. Watts, "What Can Search Predict ?" *WWW '10*, 2010.
- [3] X. Yu, Y. Liu, X. Huang, and A. An, "Mining Online Reviews for Predicting Sales Performance: A Case Study in the Movie Domain," *IEEE Transactions on Knowledge and Data Engineering*,

- vol. 24, no. 4, pp. 720–734, Apr. 2012. [Online]. Available: <http://ieeexplore.ieee.org/lpdocs/epic03/wrapper.htm?arnumber=5677530>
- [4] G. Kulkarni, P. K. Kannan, and W. Moe, “Using online search data to forecast new product sales,” *Decision Support Systems*, vol. 52, no. 3, pp. 604–611, 2012. [Online]. Available: <http://dx.doi.org/10.1016/j.dss.2011.10.017>
 - [5] L. Zhang, J. Luo, and S. Yang, “Forecasting box office revenue of movies with BP neural network,” *Expert Systems with Applications*, vol. 36, no. 3, pp. 6580–6587, Apr. 2009. [Online]. Available: <http://linkinghub.elsevier.com/retrieve/pii/S095741740800496X>
 - [6] K. J. Lee and W. Chang, “Bayesian belief network for box-office performance: A case study on Korean movies,” *Expert Systems with Applications*, vol. 36, no. 1, pp. 280–291, Jan. 2009. [Online]. Available: <http://linkinghub.elsevier.com/retrieve/pii/S0957417407004228>
 - [7] MPAA. 2012 theatrical statistics summary. [Online]. Available: <http://www.mpa.org/resources/3037b7a4-58a2-4109-8012-58fca3abdf1b.pdf>
 - [8] R. Sharda and D. Delen, “Predicting box-office success of motion pictures with neural networks,” *Expert Systems with Applications*, vol. 30, no. 2, pp. 243–254, Feb. 2006. [Online]. Available: <http://linkinghub.elsevier.com/retrieve/pii/S0957417405001399>
 - [9] E. V. Karniouchina, “Impact of star and movie buzz on motion picture distribution and box office revenue,” *International Journal of Research in Marketing*, vol. 28, no. 1, pp. 62–74, Mar. 2011. [Online]. Available: <http://linkinghub.elsevier.com/retrieve/pii/S0167811610000881>
 - [10] H. Rui, Y. Liu, and A. Whinston, “Whose and what chatter matters? The effect of tweets on movie sales,” *Decision Support Systems*, vol. 55, no. 4, pp. 863–870, Nov. 2013. [Online]. Available: <http://linkinghub.elsevier.com/retrieve/pii/S0167923612003880>
 - [11] J. Du, H. Xu, and X. Huang, “Box office prediction based on microblog,” *Expert Systems with Applications*, vol. 41, no. 4, pp. 1680–1689, Mar. 2014. [Online]. Available: <http://linkinghub.elsevier.com/retrieve/pii/S0957417413006866>
 - [12] S. Moon, P. K. Bergey, and D. Iacobucci, “Dynamic effects among movie ratings, movie revenues, and viewer satisfaction,” *Journal of Marketing*, vol. 74, no. 1, pp. 108–121, 2010.
 - [13] H. Drucker, C. J. Burges, L. Kaufman, A. Smola, and V. Vapnik, “Support vector regression machines,” *Advances in neural information processing systems*, vol. 9, pp. 155–161, 1997.
 - [14] I. Basheer and M. Hajmeer, “Artificial neural networks: fundamentals, computing, design, and application,” *Journal of microbiological methods*, vol. 43, no. 1, pp. 3–31, 2000.